

Scalability & Future Plan

Date	
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses local JSON files instead of a database for storing history and feedback.	Limits scalability and efficient data management for multiple users.	High
2	Application is designed for local execution and is not deployed on a cloud platform.	Remote access and real-world usage are not available.	High
3	GPT-2 and DistilBERT models require noticeable time to load during the first execution.	Increases initial startup time and affects user experience.	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a single/local user environment.	Deploy the application on a cloud platform and support multiple concurrent users.
2	Data Storage	Conversation history and feedback stored in JSON files.	Replace JSON storage with PostgreSQL or SQLite for efficient data management.
3	Performance	AI models are loaded locally during application startup.	Optimize model loading, cache models, or use hosted inference services for faster responses.
4	Security	Basic local execution with minimal security controls.	Implement user authentication, HTTPS, API security, and secure environment variable management.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on a cloud platform (Render/Railway/Azure).	Short Term	Enable online access and remote usage.
Phase 2	Integrate a relational database (PostgreSQL/SQLite) for persistent storage.	Medium Term	Improve scalability, reliability, and data management.
Phase 3	Integrate Google Gemini API or other advanced LLMs for improved conversation generation.	Medium Term	Produce more natural, personalized, and context-aware conversation starters.
Phase 4	Add user authentication, user profiles, analytics dashboard, and multi-user support.	Long Term	Enhance security, personalization, and enterprise-level usability.